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Anatomia de uma app Android

PDM - Programação para Dispositivos Móveis

Paulo Pereira

paulo.pereira@isel.pt

O pacote distribuível

- Build realizado com  **Gradle**
- Empacotamento é feito num ficheiro APK (Android Package) contendo
 - Código compilado (Dalvik VM)
 - Manifesto: AndroidManifest.xml
 - Recursos da aplicação: texto, imagens, fontes, etc.



Build

Android

- app
 - manifests
 - AndroidManifest.xml
 - kotlin+java
 - prodigi.pdm.demos.helloandroid
 - ui.theme
 - MainActivity
 - prodigi.pdm.demos.helloandroid (androidTest)
 - prodigi.pdm.demos.helloandroid (test)
 - res
 - drawable
 - mipmap
 - values
 - xml
 - Gradle Scripts
 - build.gradle.kts (Project: HelloAndroid)
 - build.gradle.kts (Module :app)
 - proguard-rules.pro (ProGuard Rules for ":app")
 - gradle.properties (Project Properties)
 - gradle-wrapper.properties (Gradle Version)
 - libs.versions.toml (Version Catalog "libs")
 - local.properties (SDK Location)
 - settings.gradle.kts (Project Settings)

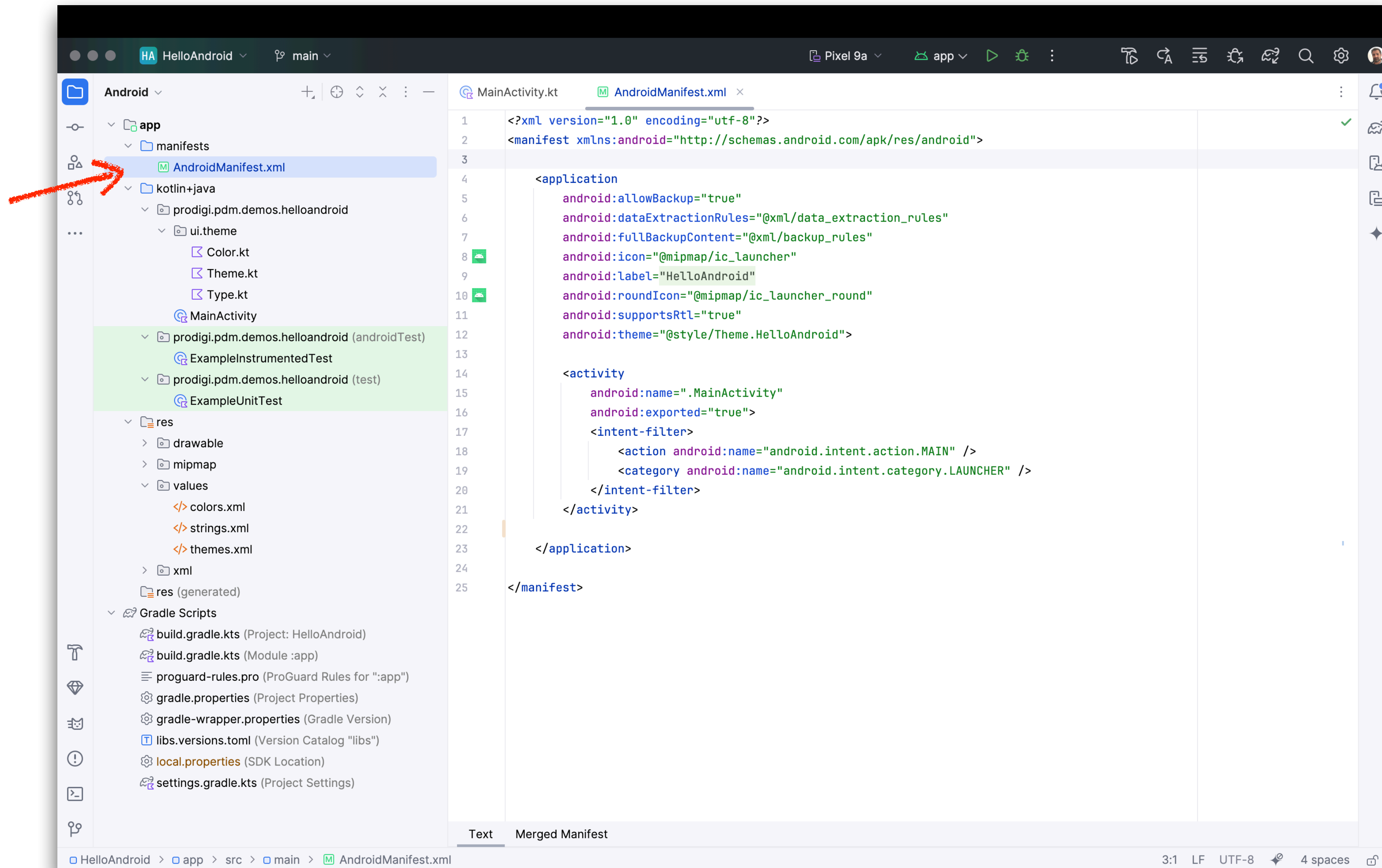
build.gradle.kts (:app)

You can use the Project Structure dialog to view and edit your project configuration [Open \(⌘;\)](#) [Hide notification](#)

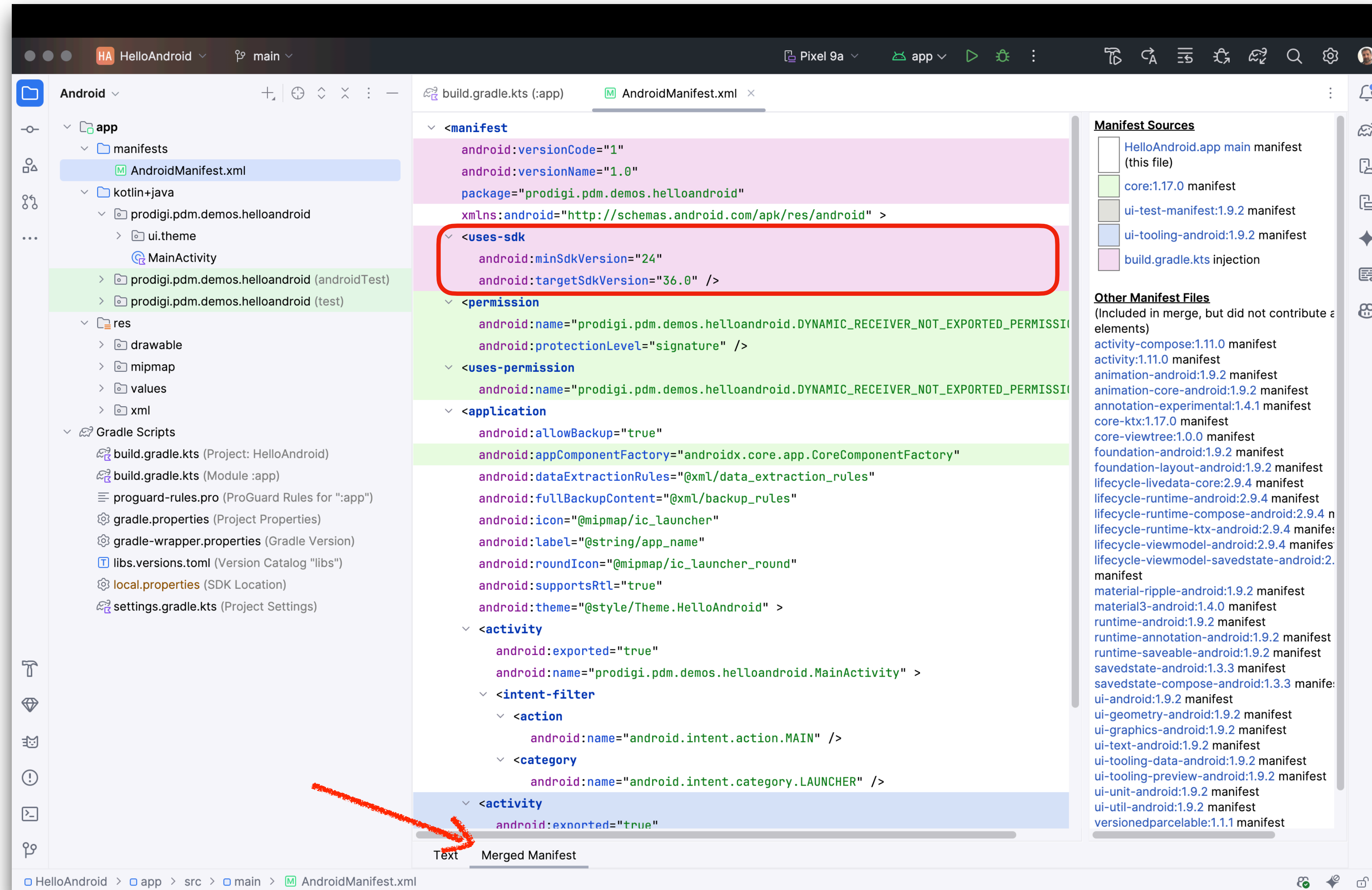
```
1 import org.jetbrains.kotlin.gradle.dsl.JvmTarget
2
3 plugins {
4     alias(libs.plugins.android.application)
5     alias(libs.plugins.kotlin.android)
6     alias(libs.plugins.kotlin.compose)
7 }
8
9 android {
10     namespace = "prodigi.pdm.demos.helloandroid"
11     compileSdk = 36
12
13     defaultConfig {
14         applicationId = "prodigi.pdm.demos.helloandroid"
15         minSdk = 24
16         targetSdk = 36
17         versionCode = 1
18         versionName = "1.0"
19
20         testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
21     }
22
23     buildTypes { ... }
24     compileOptions { ... }
25     kotlin.compilerOptions { ... }
26     buildFeatures { ... }
27 }
28
29 dependencies {
30     implementation(libs.androidx.core.ktx)
31     implementation(libs.androidx.lifecycle.runtime.ktx)
32     implementation(libs.androidx.activity.compose)
33     implementation(platform(dependencyProvider(libs.androidx.compose.bom)))
34     implementation(libs.androidx.ui)
35     implementation(libs.androidx.ui.graphics)
36 }
```



Manifesto (1)



Manifesto [2]



AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest
    android:versionCode="1"
    android:versionName="1.0"
    package="prodigi.pdm.demos.helloandroid"
    xmlns:android="http://schemas.android.com/apk/res/android" >

    <uses-sdk
        android:minSdkVersion="24"
        android:targetSdkVersion="36.0" />

    <permission
        android:name="prodigi.pdm.demos.helloandroid.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION"
        android:protectionLevel="signature" />

    <uses-permission
        android:name="prodigi.pdm.demos.helloandroid.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION" />

    <application
        android:allowBackup="true"
        android:appComponentFactory="androidx.core.app.CoreComponentFactory"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.HelloAndroid" >

        <activity
            android:exported="true"
            android:name="prodigi.pdm.demos.helloandroid.MainActivity" >

            <intent-filter>
                <action
                    android:name="android.intent.action.MAIN" />

                <category
                    android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

            <activity
                android:exported="true" />
        </activity>
    </application>
</manifest>
```

Manifest Sources

- HelloAndroid.app main manifest (this file)
- core:1.17.0 manifest
- ui-test-manifest:1.9.2 manifest
- ui-tooling-android:1.9.2 manifest
- build.gradle.kts injection

Other Manifest Files

(Included in merge, but did not contribute elements)

- activity-compose:1.11.0 manifest
- activity:1.11.0 manifest
- animation-android:1.9.2 manifest
- animation-core-android:1.9.2 manifest
- annotation-experimental:1.4.1 manifest
- core-ktx:1.17.0 manifest
- core-viewtree:1.0.0 manifest
- foundation-android:1.9.2 manifest
- foundation-layout-android:1.9.2 manifest
- lifecycle-livedata-core:2.9.4 manifest
- lifecycle-runtime-android:2.9.4 manifest
- lifecycle-runtime-compose-android:2.9.4 manifest
- lifecycle-runtime-ktx-android:2.9.4 manifest
- lifecycle-viewmodel-android:2.9.4 manifest
- lifecycle-viewmodel-savedstate-android:2.9.4 manifest
- material-ripple-android:1.9.2 manifest
- material3-android:1.4.0 manifest
- runtime-android:1.9.2 manifest
- runtime-annotation-android:1.9.2 manifest
- runtime-saveable-android:1.9.2 manifest
- savedstate-android:1.3.3 manifest
- savedstate-compose-android:1.3.3 manifest
- ui-android:1.9.2 manifest
- ui-geometry-android:1.9.2 manifest
- ui-graphics-android:1.9.2 manifest
- ui-text-android:1.9.2 manifest
- ui-tooling-data-android:1.9.2 manifest
- ui-tooling-preview-android:1.9.2 manifest
- ui-unit-android:1.9.2 manifest
- ui-util-android:1.9.2 manifest
- versionedparcelable:1.1.1 manifest

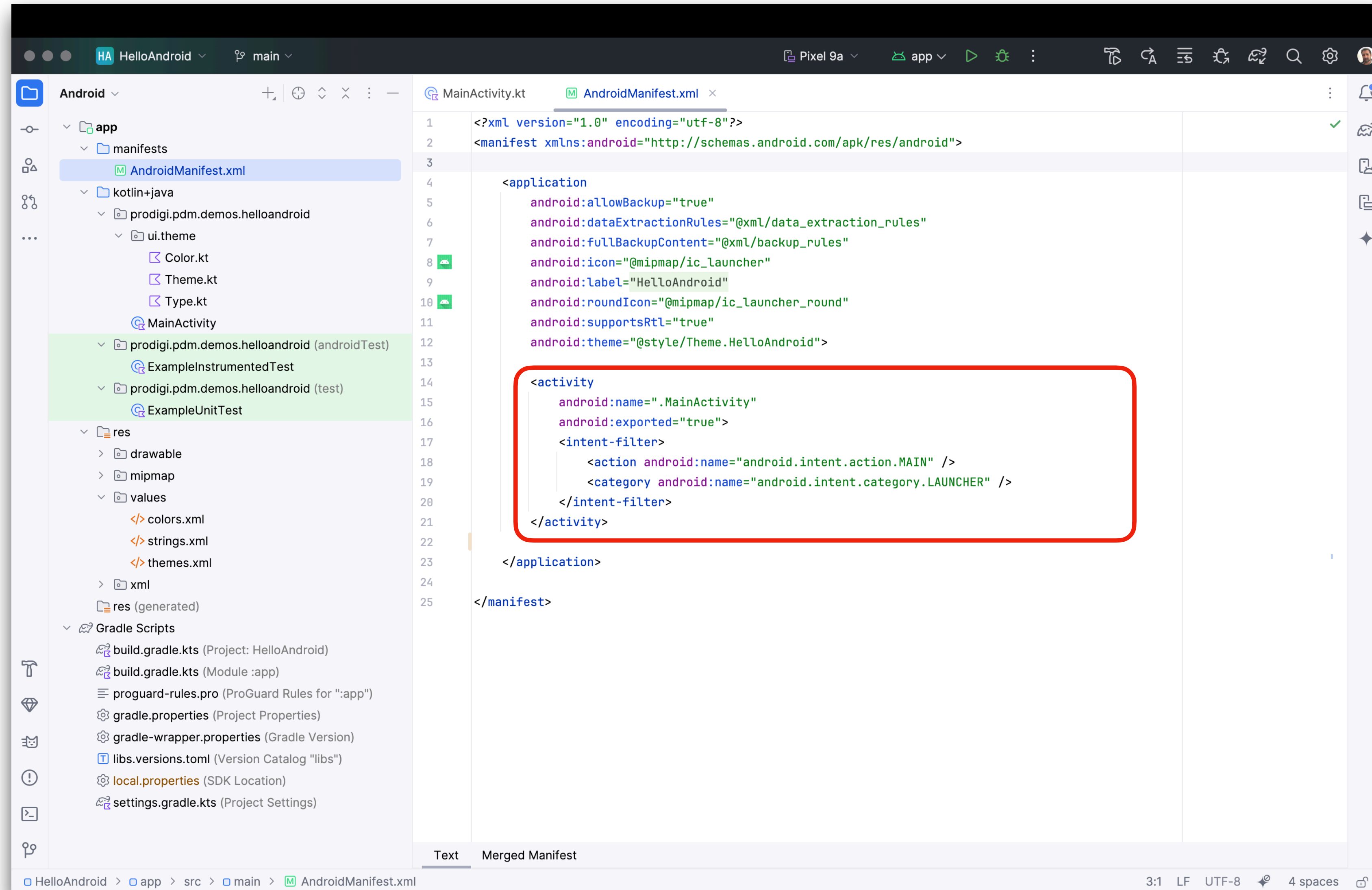


Componentes Android

- Elementos potencialmente acessíveis do exterior (i.e. *entry points*)
 - Activity ←
 - Service
 - Broadcast Receiver
 - Content Provider



Manifesto (3)



Recursos

The screenshot displays the Android Studio IDE. On the left, the 'Project' view shows the 'res' directory expanded, with a red bracket and a red circle labeled '1' highlighting the 'strings' folder. The 'strings' folder contains 'strings.xml' and 'strings.xml (pt)'. The 'strings.xml (pt)' file is selected. On the right, the 'Editor' view shows the 'strings.xml' file. The file content is as follows:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
  <string name="app_name">HelloAndroid</string>
  <string name="greeting">Olá Android</string>
</resources>
```

A red circle labeled '2' is placed next to the 'greeting' string. The status bar at the bottom shows the file path: 'HelloAndroid > app > src > main > res > values > strings.xml'.



Activity - Primeiro olhar

```
1 class MainActivity : ComponentActivity() {  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        enableEdgeToEdge()  
2 setContent {  
        HelloAndroidTheme {  
            Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->  
                3 Text(  
                    4 text = stringResource(id = R.string.greeting),  
                    modifier = Modifier.padding(paddingValues = innerPadding)  
                )  
            }  
        }  
    } // setContent ends  
} // onCreate ends  
} // class ends
```



A saber (ou rever)

- Classes - <https://kotlinlang.org/docs/classes.html>
- Interfaces - <https://kotlinlang.org/docs/interfaces.html>
- Herança e polimorfismo - <https://kotlinlang.org/docs/inheritance.html>
- *Companion objects* - <https://kotlinlang.org/docs/object-declarations.html>
- Funções de extensão - <https://kotlinlang.org/docs/extensions.html>
- HOFs e lambdas - <https://kotlinlang.org/docs/lambdas.html>





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